Math 110 — Week 7

Solutions

HW7 P1.

Idea. $A^2 = -I$ and $(B-I)^2 = 0$ give polynomials of degree 2 that are 0 at A and B; neither matrix is a multiple of I, which rules out degree 1.

1. (a) $A^2 = -I$, so $m_A \mid x^2+1$ (Prop. 65). $Ae_1 = e_2$ is not a multiple of e_1, so A is not a multiple of I and $\deg m_A \neq 1$ (F2). Hence $m_A = x^2+1$ over $\mathbb{R}$, and over $\mathbb{C}$ it is the same polynomial (F8): $x^2+1 = (x-i)(x+i)$.
2. (b) $(B-I)^2 = \left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]^2 = 0$ and $B \neq I$, so $m_B = (x-1)^2$.
3. (c) Over $\mathbb{R}$, x^2+1 has no real root, so it is not a product of real linear factors: A is not diagonalizable over $\mathbb{R}$ (Prop. 68). Over $\mathbb{C}$ its roots $\pm i$ are distinct: A is diagonalizable. m_B has the repeated root 1: B is not diagonalizable. This matches Homework 6.

HW7 P2.

Idea. D lowers the degree, so some power of D is 0; find the first one.

1. (a) $D^4 p = 0$ for every $p \in \mathcal{P}_3(\mathbb{R})$, since four derivatives of a polynomial of degree ≤ 3 vanish. So $m_D \mid x^4$, and $m_D = x^k$ for some $k \leq 4$ (divisors of a product of linear factors).
2. $D^3 x^3 = 6 \neq 0$, so $k = 4$: $m_D = x^4$.
3. (b) m_D has a repeated root, so D is not diagonalizable (Prop. 68). (Also $E(0, D)$ is the constants, of dimension $1 < 4$.)

HW7 P3.

Idea. If $a_0 = 0$, the factor T in $m_T(T) = Tq(T)$ can be cancelled because T is invertible, leaving a smaller polynomial that is 0 at T. Otherwise move the constant term to one side and factor T out of the rest.

1. (a) Suppose $a_0 = 0$. Then $m_T = xq$ with $\deg q = d-1$, and $0 = m_T(T) = Tq(T)$. Multiply by T^{-1}: $q(T) = 0$, which contradicts the minimality of m_T (if $d = 1$, $q = 1$ and $q(T) = I \neq 0$).
2. (b) From $m_T(T) = 0$,

$$T\big(T^{d-1} + a_{d-1}T^{d-2} + \cdots + a_1 I\big) = -a_0 I, \qquad \text{so} \qquad T^{-1} = -\frac{1}{a_0}\big(T^{d-1} + a_{d-1}T^{d-2} + \cdots + a_1 I\big).$$

3. (c) Here $a_0 = 4$, $a_1 = -5$, $a_2 = -1$, $a_3 = 1$, so $T^{-1} = -\frac{1}{4}(T^3 + T^2 - T - 5I)$.
4. $m_T(1) = 1+1-1-5+4 = 0$ and $m_T'(1) = 4+3-2-5 = 0$, so $(x-1)^2 \mid m_T$ (F10(a)). Dividing,

$$m_T = (x-1)^2(x^2+3x+4).$$

5. The eigenvalues are the roots (Prop. 66): 1 and $\frac{-3 \pm i\sqrt{7}}{2}$. The root 1 is repeated, so T is not diagonalizable (Prop. 68).

HW7 P4.

Idea. Substituting $x \mapsto x + \lambda$ (or $x \mapsto x/c$) turns polynomials that are 0 at T into polynomials that are 0 at $T - \lambda I$ (or cT), and back. So the two minimal polynomials have the same degree, and one divides the other.

1. (a) $p(x) = m_T(x + \lambda)$ is monic of degree $d = \deg m_T$, and $p(T - \lambda I) = m_T(T) = 0$. So $m_{T-\lambda I} \mid p$ and $\deg m_{T-\lambda I} \leq d$.
2. $r(x) = m_{T-\lambda I}(x - \lambda)$ is monic, and $r(T) = m_{T-\lambda I}(T - \lambda I) = 0$, so $d \leq \deg r = \deg m_{T-\lambda I}$.
3. A monic divisor of p of the same degree is p: $m_{T-\lambda I} = p$.
4. (b) $p(x) = c^d m_T(x/c)$ is monic of degree d (leading coefficient $c^d \cdot c^{-d} = 1$), and $p(cT) = c^d m_T(T) = 0$, so $m_{cT} \mid p$.
5. Applying this with cT and $1/c$ in place of T and c gives $d \leq \deg m_{cT}$. So $m_{cT} = p$.

HW7 P5.

Idea. The roots of m_A are the eigenvalues and $\deg m_A \leq 2$, which leaves two candidates.

1. $\deg m_A \leq 2$ (Prop. 67), and every root of m_A is an eigenvalue (Prop. 66), so λ is its only root.
2. Over $\mathbb{C}$, m_A factors into linear factors (T6), so $m_A = x - \lambda$ or $(x - \lambda)^2$.
3. $m_A = x - \lambda$ exactly when $A - \lambda I = 0$ (F2).
4. By Prop. 68, A is diagonalizable exactly when m_A has no repeated root, that is, $m_A = x - \lambda$, that is, $A = \lambda I$.

HW7 P6.

Idea. C is triangular, so the eigenvalues are on the diagonal; list the candidate minimal polynomials and test them in order of degree.

1. The eigenvalues are 2 and 3 (Prop. 60). The eigenspaces are

$$E(2) = \operatorname{null} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \operatorname{span}(e_1), \qquad E(3) = \operatorname{null} \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \operatorname{span}(e_3).$$

2. Candidates (roots exactly $2, 3$, degree ≤ 3): $(x-2)(x-3)$, $(x-2)^2(x-3)$, $(x-2)(x-3)^2$.
3. $(C - 2I)(C - 3I)$ has $(1, 2)$ entry $-1 \neq 0$. $(C - 2I)^2 = \operatorname{diag}(0, 0, 1)$, and $(C - 2I)^2(C - 3I) = 0$.
4. So $m_C = (x-2)^2(x-3)$, which has a repeated root: C is not diagonalizable. (Also $\dim E(2) + \dim E(3) = 2 < 3$.)

HW7 P7.

Idea. Lagrange interpolation: a product of factors $x - \lambda_k$ vanishes at every λ_k but λ_j; divide by its value at λ_j.

1. Let

$$p_j(x) = \prod_{k \neq j} \frac{x - \lambda_k}{\lambda_j - \lambda_k}.$$

   Then $p_j(\lambda_j) = 1$ and $p_j(\lambda_k) = 0$ for $k \neq j$.
2. Since m_T has distinct roots, T is diagonalizable and $V = \bigoplus_k E(\lambda_k, T)$ (Prop. 68).
3. For $v \in E(\lambda_k, T)$, $p_j(T)v = p_j(\lambda_k)v$, which is v if $k = j$ and 0 otherwise. So for $v = \sum_k v_k$, $p_j(T)v = v_j$: the projection onto $E(\lambda_j, T)$ along $\bigoplus_{k \neq j} E(\lambda_k, T)$.

M1.

Idea. Find a low-degree polynomial that is 0 at the matrix, then rule out its proper divisors. In (c), B is block diagonal, so m_B is the lcm of the blocks' minimal polynomials (F4).

1. (a) Every entry of A^2 is 3, so $A^2 = 3A$ and $m_A \mid x(x-3)$. $A \neq 0$ and $A \neq 3I$, so $m_A = x(x-3)$. Distinct roots: diagonalizable.
2. (b) $Pe_1 = e_2$, $Pe_2 = e_3$, $Pe_3 = e_1$, so $P^3 = I$ and $m_P \mid x^3 - 1$. Since e_1, Pe_1, P^2e_1 is a basis, no nonzero polynomial of degree ≤ 2 is 0 at P (F7(c)), so $m_P = x^3 - 1 = (x-1)(x^2+x+1)$.
3. Over $\mathbb{C}$ the roots 1 and $\frac{-1 \pm i\sqrt{3}}{2}$ (the roots of x^2+x+1) are distinct: diagonalizable. Over $\mathbb{R}$, x^2+x+1 has no real root: not diagonalizable (F8).
4. (c) B is block diagonal with blocks $B_1 = \left[\begin{smallmatrix} 3 & 1 \\ 0 & 3 \end{smallmatrix}\right]$, (3), (5). $B_1 \neq 3I$ and $(B_1 - 3I)^2 = 0$, so $m_{B_1} = (x-3)^2$; the other blocks give $x-3$ and $x-5$. By F4 and the lcm formula,

$$m_B = \operatorname{lcm}\big((x-3)^2, x-3, x-5\big) = (x-3)^2(x-5).$$

   It has a repeated root: not diagonalizable.

M2.

Idea. (a) Test the basis vectors: each x^k is an eigenvector. (b) $S - I$ lowers the degree, like differentiation, so a power of it is 0; find the first power that is 0.

1. (a) $T(x^k) = x \cdot kx^{k-1} = kx^k$, so $1, x, \ldots, x^n$ is an eigenbasis with the distinct eigenvalues $0, 1, \ldots, n$. T is diagonalizable and $m_T = x(x-1)\cdots(x-n)$.
2. (b) $(S-I)p = p(x+1) - p(x)$: the leading terms cancel, so the degree drops by at least 1, and constants go to 0. Hence $(S-I)^{n+1} = 0$, and $m_S \mid (x-1)^{n+1}$.
3. The binomial theorem gives

$$(S-I)x^k = (x+1)^k - x^k = kx^{k-1} + (\text{terms of lower degree}),$$

   so $(S-I)^n x^n = n! \neq 0$. So $m_S = (x-1)^{n+1}$.
4. It has a repeated root (as $n \geq 1$): S is not diagonalizable.

M3.

Idea. $T^3 = T$ says that $x^3 - x$ is 0 at T; factor it.

1. $m_T \mid x^3 - x = x(x-1)(x+1)$ (Prop. 65), so m_T is a product of some of these distinct real linear factors (divisors of a product of linear factors).
2. By Prop. 68, T is diagonalizable. Its eigenvalues are roots of m_T (Prop. 66), hence in $\{-1, 0, 1\}$.

M4.

Idea. A monic polynomial of degree 1 is $x - c$, and $m_T = x - c$ says $T - cI = 0$.

1. If $T = cI$, then $x - c$ is 0 at T, and no nonzero constant is ($aI \neq 0$ as $V \neq \{0\}$), so $m_T = x - c$.
2. Conversely, if $m_T = x - c$, then $T - cI = m_T(T) = 0$.

M5.

Idea. Substitute $x \mapsto 1/x$ in $m_T(T) = 0$ and clear denominators by multiplying by T^{-d}.

1. Let
$$q(x) = \frac{1}{a_0} x^d m_T(1/x) = \frac{1}{a_0}\left(1 + a_{d-1}x + \cdots + a_0 x^d\right).$$
It is monic of degree d ($a_0 \neq 0$ by HW7 P3).
2. $q(T^{-1}) = \frac{1}{a_0} T^{-d} m_T(T) = 0$, so $m_{T^{-1}} \mid q$ and $\deg m_{T^{-1}} \leq d$.
3. Applying this to T^{-1} in place of T gives $d \leq \deg m_{T^{-1}}$. So $m_{T^{-1}} = q$.

M6.

Idea. $m_A(A) = 0$ is a rule for rewriting A^2 in terms of A and I; use it to reduce any power, and solve it for A^{-1}.

1. A is not a multiple of I, so $m_A = x^2 - (\operatorname{tr} A)x + \det A = x^2 - 5x - 2$ (Method 6).
2. From $A^2 - 5A = 2I$: $A(A - 5I) = 2I$, so
$$A^{-1} = \frac{1}{2}(A - 5I) = \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix}.$$
3. Multiply $A^2 = 5A + 2I$ by A and replace A^2 again:
$$A^3 = 5A^2 + 2A = 5(5A + 2I) + 2A = 27A + 10I.$$

R1.

Idea. The two null spaces give the eigenvectors. Testing $(x-2)(x+1)$ decides whether one factor per eigenvalue is enough; $(x-2)^2(x+1)$ is the next candidate.

1. (a) The matrix is triangular, so the eigenvalues are 2 and -1 (Prop. 60). For $A - 2I$ the equations are $y = 0$ and $-3z = 0$, so $E(2, T) = \operatorname{span}(e_1)$. For $A + I$ they are $3x + y = 0$ and $3y = 0$, so $E(-1, T) = \operatorname{span}(e_3)$.
2. (b) m_A has both roots (Prop. 66), and $\deg m_A \leq 3$. The product
$$(A - 2I)(A + I) = \begin{pmatrix} 0 & 3 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \neq 0,$$
but $(A-2I)^2(A+I) = 0$. The other degree-3 candidate, $(x-2)(x+1)^2$, is not needed: m_A divides $(x-2)^2(x+1)$, and the only proper divisor containing both roots is $(x-2)(x+1)$, which fails. So $m_A = (x-2)^2(x+1)$.
3. (c) m_A has a repeated root, so T is not diagonalizable (Prop. 68). Also the eigenspaces have dimensions $1 + 1 < 3$.

R2.

Idea. The range of $T - \lambda I$ is smaller than V and contains $Tv - \lambda v$ for every v, so it controls the extra columns when a basis is extended.

1. (a) In dimension 1, any basis vector u has $Tu = au$ for some a, a 1×1 upper-triangular matrix.
2. (b) For $n > 1$, choose an eigenvalue λ. $T - \lambda I$ is not injective, so it is not surjective (T2), and $\dim U < n$. If $u = (T - \lambda I)v$, then $Tu = (T - \lambda I)(Tv) \in U$, so U is invariant.
3. (c) $T|_U$ is an operator on the smaller complex space U. By induction there is a basis $u_1, \ldots, u_m$ of U with $Tu_j \in \operatorname{span}(u_1, \ldots, u_j)$; if $U = \{0\}$, the basis is empty.
4. (d) Extend by $v_1, \ldots, v_r$. Then $Tv_j = (T - \lambda I)v_j + \lambda v_j \in \operatorname{span}(u_1, \ldots, u_m, v_j)$. Every basis vector is sent into the span of itself and the earlier ones, so the matrix is upper-triangular (Prop. 59).

R3.

Idea. Move the constant term of $m_T(T) = 0$ to one side. If that term is 0, an inverse would cancel the factor T and give a relation of smaller degree.

1. (a) Write $m_T = \sum_{j=0}^{d} a_j x^j$, $a_d = 1$. If $a_0 \neq 0$, then $T\big(-a_0^{-1} \sum_{j=1}^{d} a_j T^{j-1}\big) = I$. The bracket is a polynomial in T, so it commutes with T and is a two-sided inverse.
2. (b) If $a_0 = 0$, write $m_T = xq$ with $q \neq 0$ of degree $d-1$. If T were invertible, $Tq(T) = 0$ would give $q(T) = 0$, contradicting the minimality of m_T.
3. (c) $R^2(x, y) = R(-y, x) = (-x, -y)$, so $R^2 = -I$. R is not a multiple of I, so $m_R = x^2 + 1$ (F2).
4. (d) From $R^2 + I = 0$: $R(-R) = (-R)R = I$, so $R^{-1} = -R$, that is, $R^{-1}(x, y) = (y, -x)$. Check: $R(y, -x) = (x, y)$.

R4.

Idea. (a) A real eigenvalue of D must be a real root of a polynomial that is 0 at D; with only one eigenvalue, diagonalizable would mean D is a multiple of I. (b), (c) $T^2 = -I$, so an eigenvalue satisfies $\lambda^2 = -1$.

1. (a) $D(1) = 0$, $(D^2 + I)\sin x = (D^2 + I)\cos x = 0$ and $(D^2 + 4I)\sin 2x = (D^2 + 4I)\cos 2x = 0$. Polynomials in D commute, so $p(D) = D(D^2 + I)(D^2 + 4I)$ is 0 on the five spanning functions, hence on V.
2. If $Df = \lambda f$ with $f \neq 0$ and $\lambda \in \mathbb{R}$, then $p(\lambda) = \lambda(\lambda^2 + 1)(\lambda^2 + 4) = 0$, so $\lambda = 0$. And 0 is an eigenvalue, with $E(0, D) = \{f \in V : f' = 0\}$ the constants.
3. D has only the eigenvalue 0 and $D \neq 0$ (since $D \sin x = \cos x$), so D is not diagonalizable (Shortcuts of 2.2, "One eigenvalue only").
4. (b) $T^2 = -I$, so T is not a multiple of I and $m_T = x^2 + 1$; the eigenvalues are $\pm i$. $T(1, i) = (i, -1) = i(1, i)$ and $T(1, -i) = (-i, -1) = -i(1, -i)$. In the basis $(1, i), (1, -i)$ the matrix is $\mathrm{diag}(i, -i)$, which is upper-triangular.
5. (c) Over $\mathbb{R}$, $\lambda^2 = -1$ has no solution, so T has no eigenvector. An upper-triangular basis starts with an eigenvector (Prop. 59), so there is none.

R5.

Idea. The polynomials that are 0 at T on the single vector v behave like those that are 0 at T: division with remainder sends them to multiples of the one of least degree.

1. Existence: $m_T(T)v = 0$, so some monic polynomial is 0 at T on v; take one of least degree, p_v.
2. Uniqueness: if p and $\tilde{p}$ are both monic of that least degree with $p(T)v = \tilde{p}(T)v = 0$, then $p - \tilde{p}$ has smaller degree and $(p - \tilde{p})(T)v = 0$. A nonzero $p - \tilde{p}$, divided by its leading coefficient, would contradict minimality. So $p = \tilde{p}$.
3. Divisibility: divide $m_T = p_v h + r$ with $r = 0$ or $\deg r < \deg p_v$. Then $r(T)v = m_T(T)v - h(T)p_v(T)v = 0$, so $r = 0$ by minimality. So $p_v \mid m_T$.

R6.

Idea. Each statement is decided by an example from the Traps boxes, or by a one-line argument.

1. (a) True. For $T = I$, $V = E(1, I)$.
2. (b) False. The rotation of $\mathbb{R}^2$ by 90° has no eigenvector. (It is true in odd dimension, F11.)
3. (c) False. For I_2, $\dim E(1) = 2$ but $m = x - 1$ has exponent 1; for $\left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$, $\dim E(1) = 1$ but $m = (x-1)^2$ has exponent 2.
4. (d) True. I_2 and $\left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$ have the same diagonal, but $m = x - 1$ and $m = (x-1)^2$.

R7.

Idea. Tv is the range part of v and $v - Tv$ is in the null space, because $T^2 = T$. Commuting with T lets S pass through the defining equations of $\operatorname{range} T$ and $\operatorname{null} T$.

1. (a) Let $v \in \operatorname{range} T \cap \operatorname{null} T$: $v = Tw$ and $Tv = 0$. Then $v = Tw = T^2 w = Tv = 0$.
2. (b) For $v \in V$, $v = Tv + (v - Tv)$, with $Tv \in \operatorname{range} T$ and $T(v - Tv) = Tv - T^2 v = 0$.
3. (c) If $w = Tu$, then $Sw = STu = T(Su) \in \operatorname{range} T$. If $Tv = 0$, then $T(Sv) = S(Tv) = 0$. So both subspaces are invariant under S.
4. By (a) and (b), $V = \operatorname{range} T \oplus \operatorname{null} T$, so (T8) gives a block diagonal matrix for S in a basis of $\operatorname{range} T$ followed by a basis of $\operatorname{null} T$.